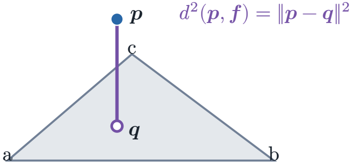
A. Exact point–triangle primitive

the closest point may lie on the face, an edge, or a vertex



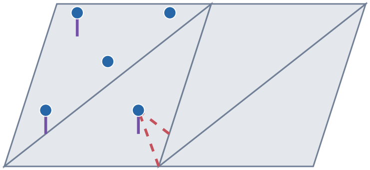
The backend evaluates squared Euclidean distance to the triangle itself, not distance to its supporting plane or centroid.

B. Two directional reductions

the same point set and triangle table induce different witness sets

solid: point → triangle

dashed: triangle → point

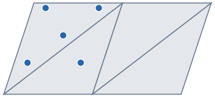


Only five representative segments are drawn; the implementation reduces over every point and every face.

C. Controlled tessellation check

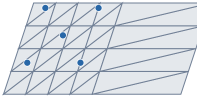
same planar support and points; only the face table changes

4 faces



$D_{M\to P}$ =0.03640 m^2

40 faces



$D_{M\to P}$ =0.02284 m^2

The equal-face completeness estimate changes because non-uniform refinement changes the number of equally weighted summands. Accuracy remains 0.00640 m^2 in both fixtures.

Controlled validity fixture computed by the repository's PyTorch3D metric and exact Trimesh closest-point witnesses; it is not an ASE performance result.